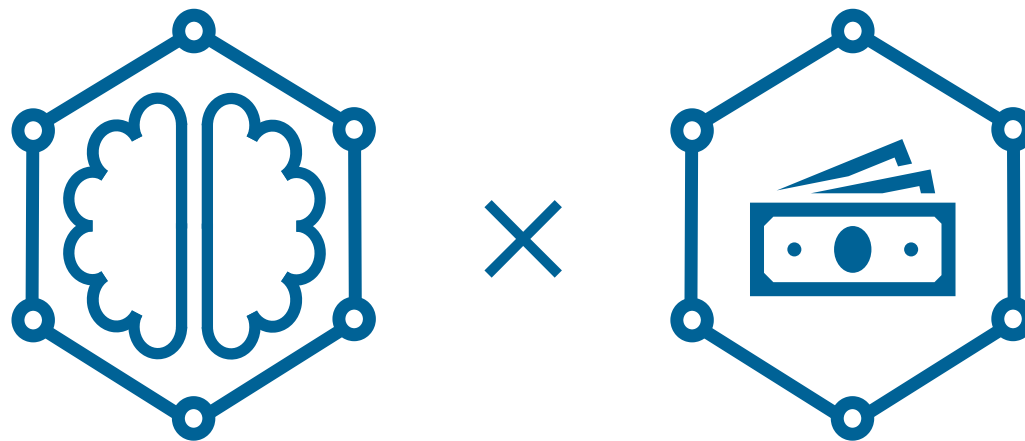




Overview

Applications

With Financial KGs as a Running Example

Emanuel Sallinger



Applications



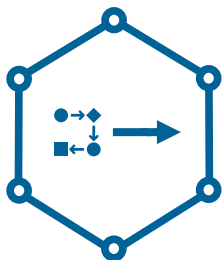
Real-World Applications

Overview of diverse applications



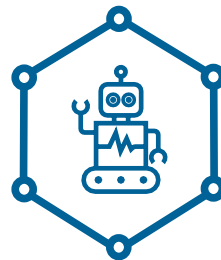
Financial KGs

Concrete applications in finance and economics



Services

Which service to provide based on KGs?



Connections

.. between KGs, AI, ML and Data Science.



Applications



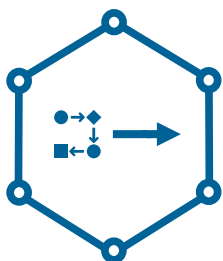
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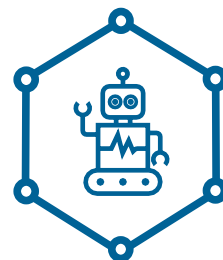
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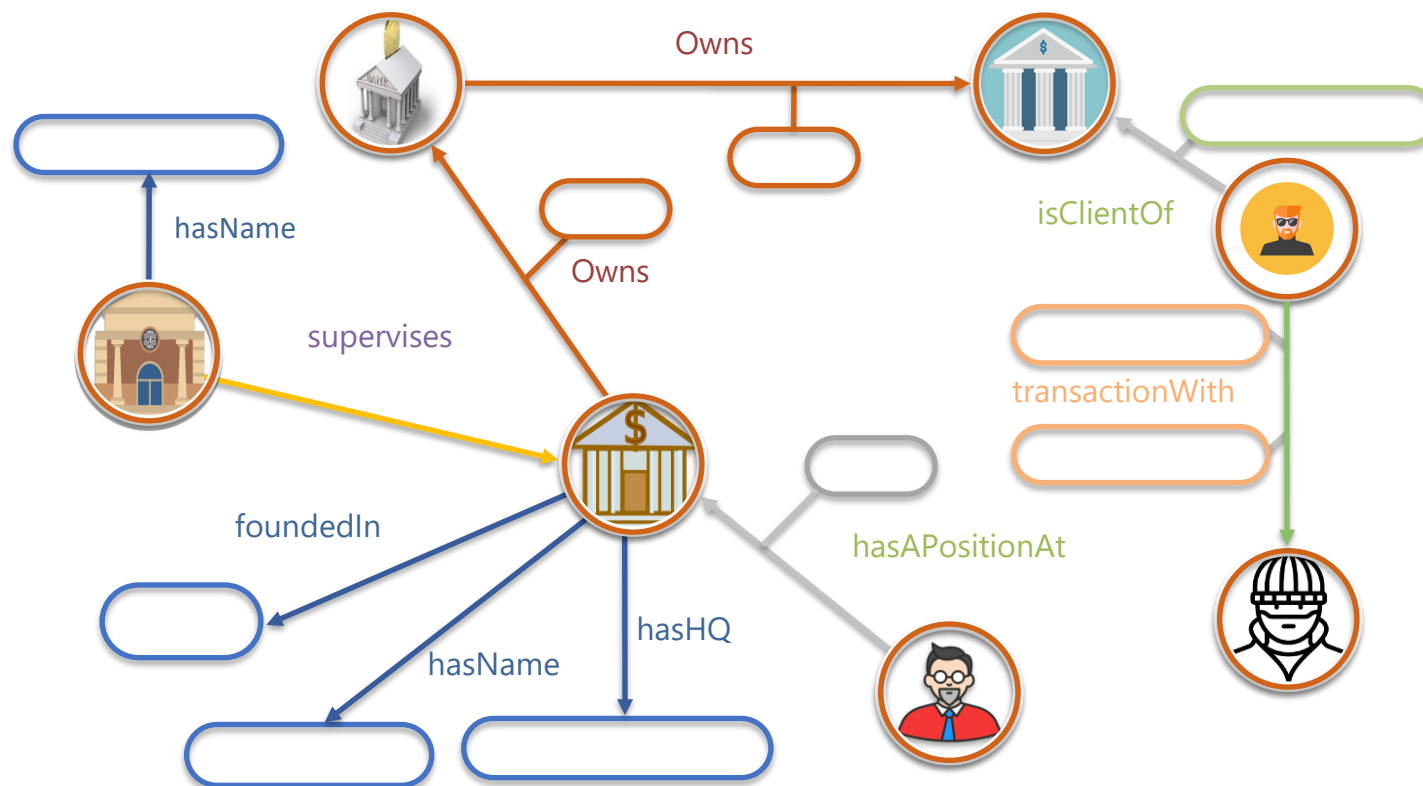
Real-World Applications

Overview of diverse applications



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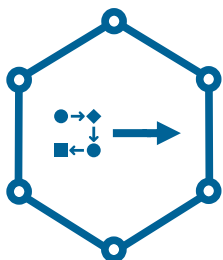
Real-World Applications

Overview of diverse applications



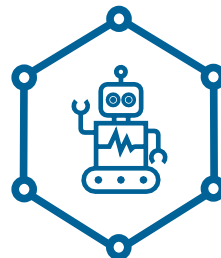
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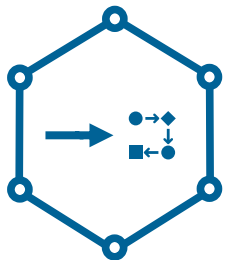
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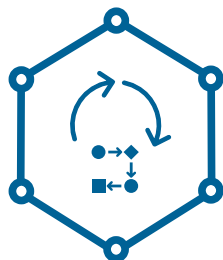
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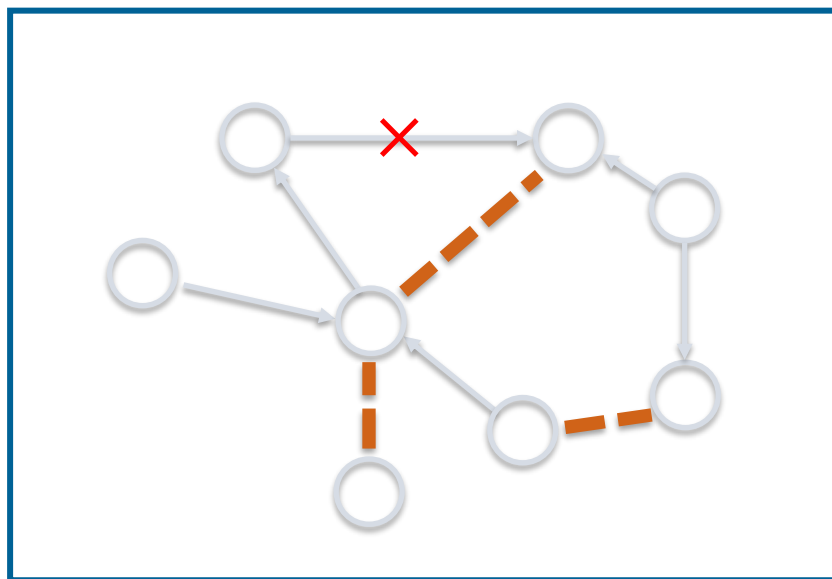
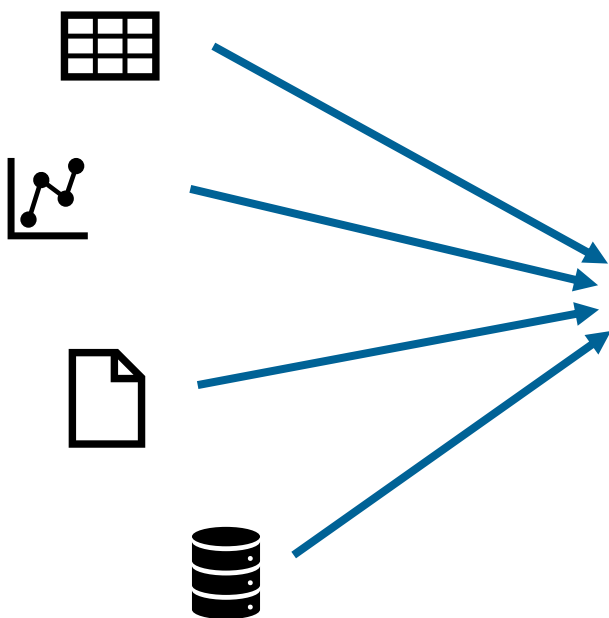
KG Creation

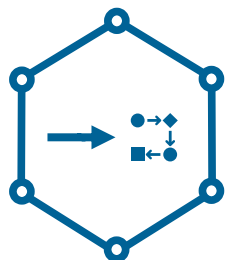
How to create a KG from heterogeneous data?



KG Evolution

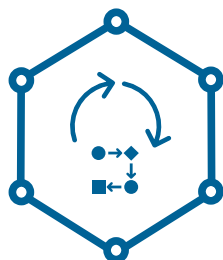
How to update, correct and complete a KG?





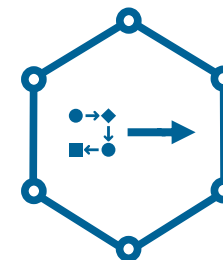
KG Creation

How to create a KG from heterogeneous data?



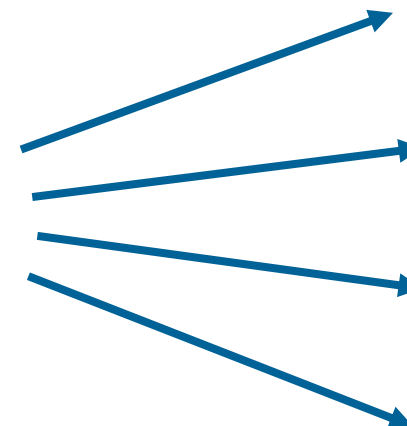
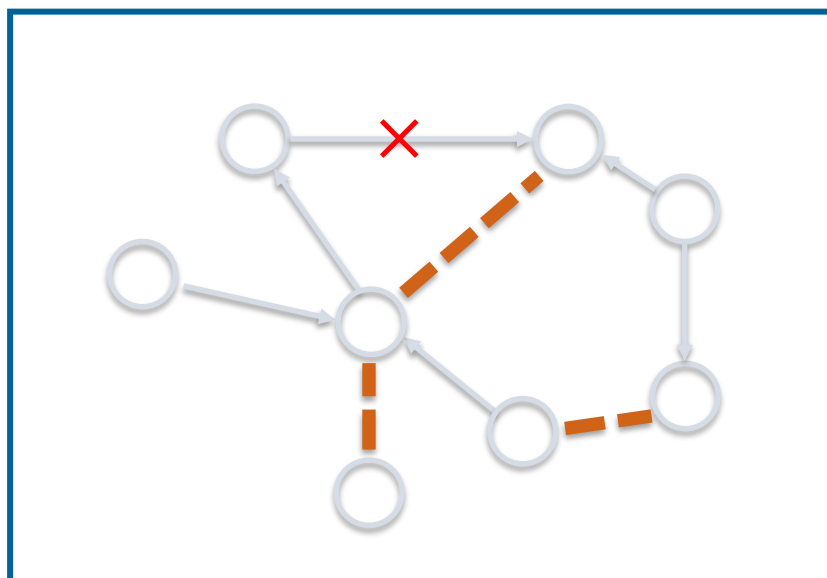
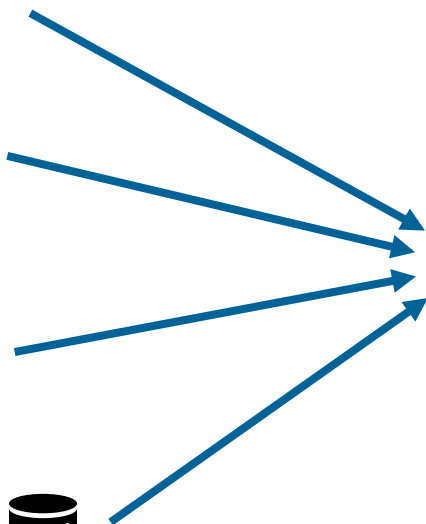
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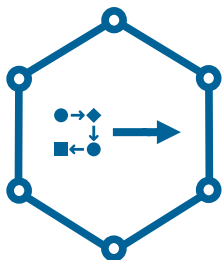
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Overview of diverse applications



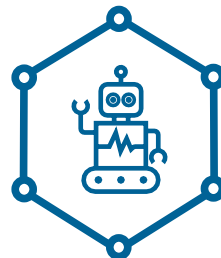
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Data Science

Artificial Intelligence

Knowledge Graphs

Machine Learning

Neural Networks

Deep Learning



Applications



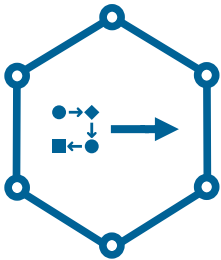
Real-World Applications

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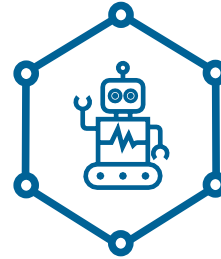
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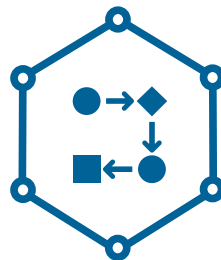
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Representations



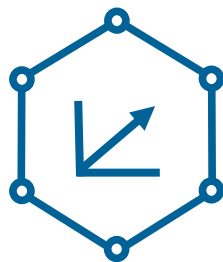
Graph



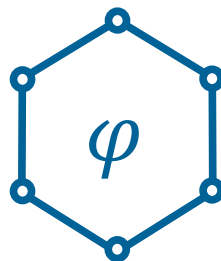
Knowledge Graph
Data Models



Knowledge



Knowledge Graph
Embeddings



Logical
Knowledge



Graph Neural
Networks



Systems



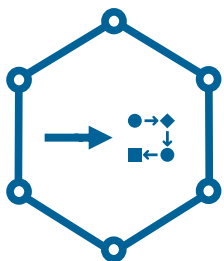
Architectures

The big picture of building IT architectures for KGs



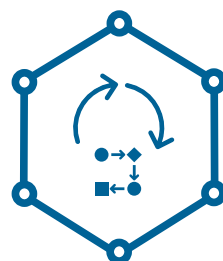
Scalable Reasoning

Making use of the knowledge in the KG



KG Creation

How to create a KG from heterogeneous data?



KG Evolution

How to update, correct and complete a KG?



Applications



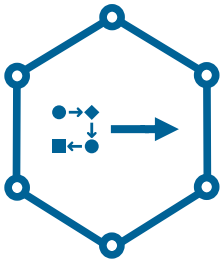
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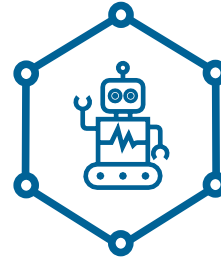
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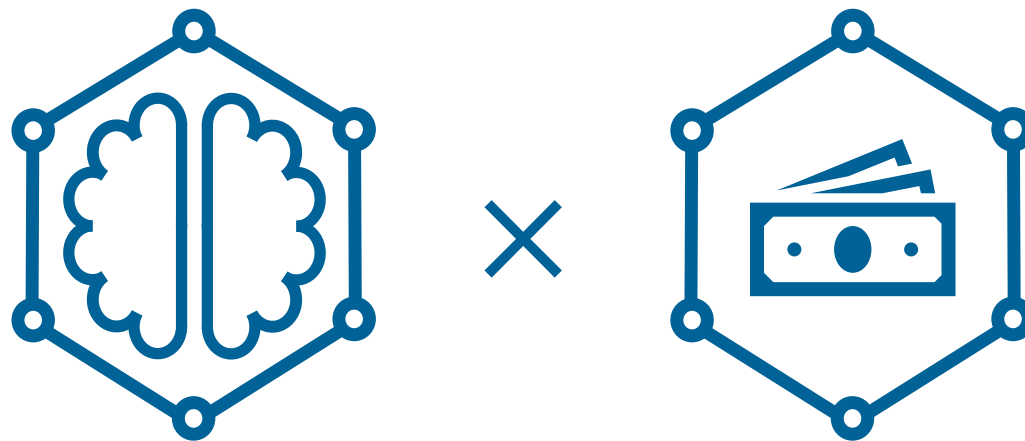
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